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Multi-tracker meta-search for qBittorrent â€” self-hosted, containerised, private-tracker-aware.

[Quick start](#) Â· [Tokens & keys](#) Â· [Documentation](#) Â· [Architecture](#) Â· [Testing](#) Â· [Contributing](#)

python tests585 passing

frontend tests182 passing

plugins48

merge serviceFastAPI :7187

ci auto (syntax / unit / integration / nightly / security)

scanners snyk | sonar | bandit | ruff | semgrep | trivy | gitleaks | pip-audit

license Apache 2.0

Features

- **Merge Search Service** â€” FastAPI service (:7187) that fans out across 40+ trackers, deduplicates results, streams via SSE.
- **Real-time results** â€” result_found events arrive as each tracker completes, no blocking.
- **Private-tracker bridge** â€” Authenticated downloads via webui-bridge.py for RuTracker, Kinozal, NNM-Club, IPTorrents.
- **Freeleech-only IPTorrents** â€” Automation never costs ratio (see [constitution VIII](#)).
- **Opt-in quality stack** â€” SonarQube + Snyk + Semgrep + Trivy + Gitleaks + bandit + pip-audit behind docker-compose.quality.yml.
- **Opt-in observability** â€” Prometheus + Grafana dashboards behind the same profile system.
- **Hook system** â€” Register scripts via POST /api/v1/hooks for search / download events.
- **Angular 21 SPA dashboard** â€” dark-themed, signals-based, per-tracker status chips with CAPTCHA re-login, virtual-scroll-ready sort.
- **PWA-ready** â€” favicon + launcher icons + web manifest ship from frontend/public/.

- **Container-first** – rootless Podman or Docker, single `./start.sh` boot.

Quick start

```
git clone https://github.com/milos85vasic/qBitTorrent.git
cd qBitTorrent

# Optional: configure tracker credentials + tokens
cp .env.example .env
$EDITOR .env          # see "Tokens & API keys" below

./setup.sh            # one-time
./start.sh            # Angular build + containers + plugin sync
```

Then visit:

URL	Purpose	Auth
http://localhost:7187/	Merge-search dashboard (SPA)	–
http://localhost:7187/docs	FastAPI Swagger UI	–
http://localhost:7187/openapi.json	OpenAPI schema	–
http://localhost:7186/	qBittorrent WebUI proxy	admin / admin
http://localhost:7185/	qBittorrent internal (container)	–
http://localhost:7188/	WebUI bridge (private-tracker downloads)	–

Tokens & API keys

– **Configure any credentials you need in `.env` before running `./start.sh`.** The single source of truth is [docs/TOKENS_AND_KEYS.md](#) – which variable is mandatory, which is optional, and **where to register** for each one.

At a glance

Category	Mandatory	Optional	Documentation
qBittorrent WebUI	–	WEBUI_USERNAME, WEBUI_PASSWORD	§1
Private trackers	per-tracker	–	§2
Public tracker uplift	–	JACKETT_API_KEY, –	§3

Category	Mandatory	Optional	Documentation
Metadata enrichment	â€”	TMDB_API_KEY, TVDB_API_KEY, â€”	Â§4
Security scanning	â€”	SNYK_TOKEN, SONAR_TOKEN, â€”	Â§5
Observability	â€”	GRAFANA_USER, GRAFANA_PASSWORD	Â§6
Orchestrator tuning	â€”	ALLOWED_ORIGINS, MAX_CONCURRENT_TRACKERS, â€”	Â§7

Where to register (fast links)

Provider	Signup	Notes
RuTracker	https://rutracker.org/forum/register.php	Username + password
Kinozal	https://kinozal.tv/signup.php	May require invite
NNM-Club	https://nnmclub.to/forum/ucp.php?mode=register	Cookie-based auth
IPTorrents	https://iptorrents.com/	Invite-only; freeleech enforced
Jackett	https://github.com/Jackett/Jackett	Self-hosted indexer
TMDb	https://www.themoviedb.org/signup	v3 auth API key
TVDb	https://thetvdb.com/api-information	Subscription API
MusicBrainz	https://musicbrainz.org/doc/MusicBrainz_API	Free â€” needs UA only
Snyk	https://app.snyk.io/	Account â†’ Auth Token
SonarCloud	https://sonarcloud.io/account/security/	Or run the compose SonarQube
Gitleaks	https://gitleaks.io/	Optional commercial license

Documentation

Everything the platform offers, indexed:

Tracked-Items + Status Documents

The mandated entry point (Â§11.4.57 / Â§11.4.212) for every tracker and status document in this repository â€” Last modified and Revision are

sourced verbatim from each document’s own revision header (Â§11.4.44); a “ means the document currently carries no revision header or has no HTML/PDF export yet (never fabricated).

Document	Last modified	Revision	Markdown	HTML
Issues	2026-06-09T20:00:00Z	5	Markdown	HTML
Issues_Summary	“	“	Markdown	HTML
Fixed	2026-06-09T21:00:00Z	13	Markdown	HTML
Fixed_Summary	“	“	Markdown	HTML
CONTINUATION	2026-06-16T23:55:00Z	19	Markdown	HTML
PORTING-FROM-LAVA	2026-07-01T16:18:43Z	1	Markdown	“ (n yet expor
REMAINING_WORK_PLAN	2026-08-07T19:10:54Z	2	Markdown	HTML
GOVERNANCE_AUDIT_2026-08-07	2026-08-07T19:10:54Z	1	Markdown	“ (n yet expor
browser_extension/Status	2026-06-13T13:10:00Z	15	Markdown	HTML
browser_extension/ Status_Summary	2026-06-10T20:05:00Z	3	Markdown	HTML
browser_extension/ RELEASE_READINESS	2026-06-13T13:10:00Z	4	Markdown	HTML
COMPLETION_STATUS	2026-06-06T00:00:00Z	1	Markdown	HTML
RELEASE_READINESS_20260616	2026-06-16T10:30:00Z	1	Markdown	HTML
features/Status	2026-06-16T23:30:00Z	7	Markdown	HTML
features/Status_Summary	2026-06-16T23:30:00Z	7	Markdown	HTML

Getting started & operation

- [docs/USER_MANUAL.md](#) “ end-user walkthrough
- [docs/TOKENS_AND_KEYS.md](#) “ a ♦ **credentials / tokens / env vars (mandatory vs optional + registration links)**
- [docs/DESIGN_SYSTEM.md](#) “ runtime-switchable palette catalogue (8 palettes, dark + light, Darcula default)
- [docs/PLUGINS.md](#) “ the 48 plugin engines
- [docs/PLUGIN_TROUBLESHOOTING.md](#) “ what to check when a plugin breaks

Architecture & subsystems

- [docs/architecture/](#) “ 5 Mermaid diagrams (topology, search lifecycle, plugin execution, bridge, shutdown)
- [docs/DATA_MODEL.md](#) “ Pydantic schemas & lifecycle (no relational DB)
- [docs/CONCURRENCY.md](#) “ asyncio semaphore + TTL caches + retry/backoff
- [docs/OBSERVABILITY.md](#) “ Prometheus + Grafana + OpenTelemetry
- [docs/PERFORMANCE.md](#) “ benchmark + load + stress test layout

- [docs/api/openapi.json](#) – frozen OpenAPI spec (diffed in CI)

Quality & security

- [docs/SECURITY.md](#) â€” threat model + credential storage
- [docs/SCANNING.md](#) â€” Snyk + Sonar + Semgrep + Trivy + Gitleaks + bandit + pip-audit
- [docs/QUALITY_STACK.md](#) â€” the opt-in docker-compose.quality.yml stack

Testing

- [docs/TESTING.md](#) â€” catalogue of every test type (30 rows)
- [docs/COVERAGE_BASELINE.md](#) â€” per-module coverage gates
- [tests/README.md](#) â€” layout of the tests/ tree

Development

- [CONTRIBUTING.md](#) â€” contribution workflow
- [CLAUDE.md](#) â€” Claude Code agent protocol (TDD + rebuild-reboot)
- [AGENTS.md](#) â€” runtime development guidance
- [.specify/memory/constitution.md](#) â€” **binding architectural contract (v1.1.0)**

Courses (self-paced, Asciiinema)

- [courses/01-operator/](#) â€” Your first search
- [courses/02-plugin-author/](#) â€” Authoring a nova3 plugin
- [courses/03-contributor/](#) â€” TDD + rebuild-reboot deep dive
- [courses/04-security-ops/](#) â€” Threat model + scanner bundle

Release / CI

- [CHANGELOG.md](#)
- [releases/README.md](#) â€” release artefacts layout
- [docs/OUT_OF_SANDBOX.md](#) â€” items requiring external credentials (HelixOA / OpenCode / submodule orgs)

Architecture

[illegible]

http://:7188 "€" webui-bridge.py (host process) "€" private-tracker b

```
Opt-in (docker-compose.quality.yml):
  http://:9000    SonarQube           profile: quality
  http://:9090    Prometheus          profile: observability
  http://:3000    Grafana             profile: observability
  (run-once)     Snyk / Semgrep /   profile: run-once
                  Trivy / Gitleaks
```

Full Mermaid renders in [docs/architecture/](#).

Testing

```
# Non-live-HTTP suites â€” run anywhere, ~10s
python3 -m pytest tests/unit/ tests/e2e/ tests/contract/ --no-cov -q

# Benchmarks â€” dedupe perf + live-HTTP fan-out benchmarks, ~4min
python3 -m pytest tests/benchmark/ --no-cov -q

# Security â€” hits the live merge service, needs stack up
python3 -m pytest tests/security/ --no-cov -q

# Integration â€” same
python3 -m pytest tests/integration/ --no-cov -q -m "not requires_credentials"

# Frontend
npx --prefix frontend ng test --watch=false

# Full scanner sweep (non-interactive; skips scanners with missing tokens)
./scripts/scan.sh --all
```

See [docs/TESTING.md](#) for the 30-row test-type catalogue.

Releases

Non-interactive builder at [scripts/build-releases.sh](#) produces artefacts under [releases/<version>/](#):

```
./scripts/build-releases.sh          # all targets, all channels
./scripts/build-releases.sh frontend # one target
./scripts/build-releases.sh --channel release
```

Each artefact ships with `SHA256SUMS + BUILD_INFO.json`.

Contributing

See [CONTRIBUTING.md](#) and the TDD protocol in [CLAUDE.md](#). PRs must keep the following green:

- Python unit + e2e + contract (pytest â€” 585+ tests)
 - Frontend Vitest (ng test â€” 182+ tests)
 - Ruff + bandit + shellcheck (via `scripts/scan.sh`)
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License

Apache 2.0 â€” see [LICENSE](#).